

Gregory Bian

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Computer Engineering (Co-op)

Expected May 2028

Vancouver, BC

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python, Java, JavaScript (React), Bash, ARM Assembly,

Firmware & Embedded Systems: FreeRTOS, STM32, CAN, SPI, I2C, UART, FPGA

Systems & Tools: Git, Linux (Ubuntu), CMake, Docker, Pytest, JUnit

WORK EXPERIENCE

Geotab

Burnaby, BC

Embedded Software Developer Intern – Vehicle Platform Development

May 2026 – Present

- Owned end-to-end firmware development for **8 new vehicle platforms** across EV, PHEV, HEV, and ICE powertrains, expanding GO device support to **10+ OEMs** including Tesla, Hyundai, Volkswagen, and Renault.
- Integrated 30+ vehicle signals and identification tags using OBD-II (SAE J1979), UDS (ISO 14229) over ISO-TP, J1939, and passive broadcast CAN, giving fleet customers EV battery, range, speed, and odometer data on newly supported vehicles.
- Developed automatic vehicle signal-processing workflows, implementing CAN parsing, logging logic, and automated validation scripts to improve the **availability of telematics data for clients** across multiple vehicle platforms.
- Leveraged Claude Skills and custom plugins to automate vehicle platform generation, vehicle signal integration, and test suite debugging, developing reusable plugins adopted by 10+ engineers to streamline workflows.
- Debugged firmware and hardware issues across vehicle platforms, analyzing CAN traffic, device logs, and hardware behavior to identify root causes and implement **reliable fixes for customer-facing vehicle integrations**.
- Authored and debugged Unit and SIL test suites, simulating real-time CAN-BUS signals and requests, validating GO device firmware logic and behavior under **time-constrained scenarios**.

ENGINEERING STUDENT DESIGN TEAMS

UBC Solar Racing

Vancouver, BC

Embedded Systems Team Lead

May 2026 – Present

- Leading a 6-person embedded team within a cross-disciplinary team of 80+ software, electrical, and mechanical engineers to design, prototype, and race a solar car, Cascadia, at FSGP and ASC solar racing competitions.
- Led end-to-end integration tests, verifying CAN communication across ECUs, debugging failing firmware logic, and testing safety features for motor control, ensuring precise and reliable control over all safety-critical systems.
- Automated the CAN processing workflow from firmware definitions to DBC generation and Grafana dashboard deployment using Claude Skills, reducing repetitive engineering effort by 70% and improving data consistency.

Embedded Firmware Developer

Sep 2025 – May 2026

- Architected firmware for the Driver Dashboard, integrating functionality for a 5-page LCD Screen, accelerator potentiometer signal processing for motor control, GPIO-based lighting controls, and CAN communication.
- Developed firmware for the vehicle telemetry board, writing low-level drivers for UART and I2C peripherals including radio, cellular, GPS, and IMU modules to enable reliable vehicle data collection and communication.

PROJECTS

NotiFlow Course Tracker | *JavaScript (React), Flask, Python*

Nov 2025 – Dec 2025

- Led a team of 5 in developing a Course Tracking Chrome Extension; implemented a React-based UI and integrated Chrome Local Storage APIs to store data locally, improving usability and reliability.
- Leveraged Flask routing to send and receive data between the Chrome Extension and backend Python scripts and support features such as Canvas API access, BeautifulSoup webscraping and OpenAI API prompt filtering.

Bare-Metal Tron Game | *C, RISC-V Assembly, FPGA*

Nov 2025

- Developed a bare-metal Tron game in C, generating VGA output for real-time graphics rendering on a DE10-Lite FPGA and supporting multiplayer or single-player gameplay against a logic-driven robot.
- Implemented hardware interrupt-driven control logic using RISC-V Assembly to handle player and robot movement, collision detection, and real-time HEX display updates.